

MediLink Summarized



MediLink AI

Bridging the “Incomplete History” Gap and making visits time-efficient

MediLink AI is a patient-controlled digital health passport that converts scattered medical records into structured medical timelines and AI-generated summaries powered by a RAG-based system, while every access and modification is cryptographically audited through blockchain with verified doctor identities.



THE PROBLEM

Healthcare decisions often rely on **fragmented and subjective patient history**, which leads to inefficiency and medical risk.

1) Patient Bias

- Patients often carry only the **records they *think* are relevant** to a specific visit, causing doctors to **miss critical background information**.

2) Memory & Communication

- History-taking is hindered by **patients forgetting details, being unable to relay** their entire history **in one sitting**, or **dismissing symptoms** due to introversion.

3) Data Isolation

- Hospitals operate on siloed systems, leaving doctors without access to past diagnoses, medications, or surgeries from other facilities.

4) Time consumption and Safety Risks

- **60–80% of consultation time is wasted** on manual history-taking, yet doctors still miss dangerous drug interactions or allergy conflicts.
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THE SOLUTION

The Secure AI Health Passport

MediLink AI creates a **portable AI-powered medical identity** that organizes a patient's entire medical history into a structured, verifiable, and secure system.



AI Medical Summary & Intelligence Engine (RAG and ML-Based)

MediLink AI uses a **Retrieval-Augmented Generation (RAG) system** to analyze patient medical records, ML models to recognise and generate structured medical insights grounded in real documents.

The AI retrieves relevant information from uploaded reports before generating responses, which helps **reduce hallucinations and ensures answers are tied to actual medical data**.

The system provides:

- **AI-generated medical summaries** based on patient records
- **Chronological diagnoses and medication timelines**
- **Allergy and risk factor identification**
- **Clinical pattern recognition across reports**

Doctors can also **interact with the AI through natural language chat**, asking questions about the patient's history.

All AI responses / summaries include **references to the specific reports, lab results, or research sources used**, ensuring transparency and reliability.



THE FEATURES

1) 📁 Lifelong Digital Medical Timeline

AI extracts data from all uploaded medical documents and builds a **structured chronological health timeline**, eliminating reliance on patient memory.

2) 🔗 FHIR Interoperability

Records are converted into **global healthcare interoperability standards (FHIR)**, bridges the gap between **isolated hospital databases**.

3) 📊 Smart Summary Dashboard with RAG and ML based summaries

Doctors scan a **secure patient QR code** or **VIEW patient dashboard** to instantly access a concise medical overview.

The dashboard shows:

- AI-generated patient summary with Risk alerts
- Diagnoses, lab tests, visits, etc timeline with NLP query engine
- Current medications with Allergies and warnings

This ensures **critical medical details are never overlooked during consultations**.

4) ⚠️ AI Risk Detection

Machine learning models automatically detect hidden clinical risks or predictions such as:

- Dangerous drug interactions
- Allergy conflicts
- Medication duplication
- Patterns indicating potential health issues

These alerts highlight risks **patients may not even know to report**.

5) 🔒 Blockchain Audit Layer

All access and modifications to patient records are logged on a **blockchain audit trail**.

Each entry records:

- Doctor ID / Phone number (an OTP is sent to confirm audit)
- Timestamp
- Access or update action

This ensures **tamper-resistant medical summaries and transparent access history**.

Impact

Time Efficiency

Recovers **50–70% of consultation time** for actual treatment instead of manual history collection.

Precision

Replaces subjective patient reporting with **verified clinical records and AI-generated summaries**.

Security & Trust

Blockchain-verified access logs ensure **data integrity and accountability**.

Clinical Confidence

Doctors receive a **complete, structured medical overview instantly**, enabling faster and safer decisions.

Core Idea in One Line

An AI-powered health passport that uses RAG and ML to summarize and analyze a patient's medical history while blockchain ensures the data cannot be tampered with.

Additional Problem it Solves

The problems current existing systems have but our app solves:

1. **AI Hallucination** in Medical Assistants & Lack of **Source Transparency**
→ **RAG system and sourcing reports**
2. ML systems need consistent and formatted clinical parameters
→ AI converts records into **FHIR-compatible standardized formats**
3. Modification tracking and patient-controlled data access
→ **Blockchain audit**
4. Fragmented patient records across hospitals
→ **Unified health passport**
5. Hidden clinical patterns across multiple reports
→ **ML risk detection**
6. Medical records stored as unstructured PDFs and scans
→ **AI document extraction**